

University of Niagara Falls Canada

Data Analytics Case Study 1

(DAMO-501-3)

Toronto Building Permits

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1. Problem Definition and Research Question

1.1 Problem Definition

The City of Toronto manages one of Canada's largest and most complex urban construction environments. Thousands of building permits are issued annually to regulate construction and ensure that development aligns with zoning policies, safety standards, and long-term urban growth strategies. Building permits are therefore not only an administrative requirement but also crucial for the growth of cities. They shape how neighbourhoods evolve and ensure that construction supports public interest goals such as housing access and safety.

Differences in approval rates, processing times, and permit types across Toronto's wards may indicate potential bias or inconsistency in how regulations are applied. Previous research on Toronto's building permitting process has revealed systemic inefficiencies and inconsistencies in approval timelines, particularly across different municipalities (Kamellia et al., 2017). These inefficiencies slow the delivery of much-needed housing and lead to significant financial losses for the city through deferred tax revenue and construction investment. The study further highlighted inconsistencies and communication gaps between City Council and the Ontario Municipal Board, suggesting that procedural inefficiencies can lead to bias and reduce public confidence in the government.

The problem this study addresses is the potential relationship between geographic location, structure type, and building permit approval outcomes within the City of Toronto. By analyzing data from the Toronto Building Permits dataset, this research aims to determine whether project location and structure characteristics significantly affect permit approval status and whether patterns exist that could indicate bias. Understanding these relationships is essential for promoting equitable and efficient urban development processes.

This investigation supports previous research emphasizing the importance of effective construction planning systems and permit processes in promoting urban growth and equitable access to housing (Rodriguez-Pose & Storper, 2020). By examining Toronto's building permit data through a data analytics lens, this study helps explain how policymakers can make evidence-based adjustments to ensure urban planning policies remain fair, consistent, and accessible across all parts of the city.

1.2 Research Questions

The study will address the following research questions:

1. Does the geographic location affect the likelihood of a building permit being approved?
2. To what extent does structure type, along with its location, influence approval outcomes?
3. Are there observable biases in permit approvals across different structure types?

2 Hypothesis Formulation

2.1 Hypothesis Set 1: Location and Permit Category

- H_0 (Null Hypothesis): There is no correlation between location and permit Approval.
- H_1 (Alternative Hypothesis): There is significant correlation between location and permit approval.

2.2 Hypothesis Set 2: Location and Permit Status

- H_0 (Null Hypothesis): There is no correlation between project location and permit status.
- H_1 (Alternative Hypothesis): There is a significant correlation between project location and permit status.

2.3 Hypothesis Set 3: Structure Type vs Permit Status

- H_0 (Null Hypothesis): There is no correlation between project structure type and permit Approval.
- H_1 (Alternative Hypothesis): There is a significant correlation between project structure type and permit approval.

3 Data Collection and SQL Queries

3.1 Data Collection Methodology

Data was collected from the Toronto Building Permits database using SQL queries that joined multiple tables such as permit, structure_type, permit_type, work_type, permit_category, and project_category. This was an effective tool that allowed for the extraction of relevant data relating to the research questions and contributed to further analysis of the impact of location and structure type on the permit approval rates. Due to the dataset having limited ward-level information, the postal codes for various areas in Toronto were used to represent the geographic location for the study, as it is a standard geographical segmental approach used for analysis in Canada.

3.2 Extracting Core Tables and Previewing Data

Before extracting analytical subsets, the core tables were inspected to verify their structure, understand available attributes, and identify relevant fields. The queries shown in Figure 3.1 ensured that the correct joins were used and that the selected variables aligned with the research question and hypothesis through inspecting the tables, confirming record counts, and observing relationships among the permit, structure_type, address, ward, and permit_status tables.

3.3 Extracting Structure Type and Permit Status

Using the permit, structure type, and permit status tables, the following queries in figure 3.2 were used to extract the number of permits within each combination of structure type and permit status from the dataset. This directly supports the investigation of whether approval outcomes vary across different structure types.

3.4 Examining Location and Permit

Using the permit, address, and permit status tables, the following queries in Figure 3.3 were used to calculate the total number of permits and the approval rate for each postal code. The data extracted from this query was later used to investigate the impact of location on the approval rate for permits.

4 Data Understanding

4.1 Overview of Key Variables

The Toronto Building Permits database contains several attributes related to project characteristics, location, and permit outcomes. The key variables outlined in the hypothesis are the location, structure type, and permit status.

Postal code is the primary indicator for location from which building permits were requested and is used as the independent variable in the first two hypotheses. It represents the area where each project is located and is used to examine if location is a determining factor for permit approvals or the distribution of permit status.

Structure type is the independent variable in the third hypothesis. It identifies the type of building or project associated with the permit and was used to investigate whether different structure categories experience different approval patterns.

Permit status is the dependent variable across all models. Project approvals cannot be represented as a simple quantifiable number, as the number of permit requests varies across postal codes. To account for this uneven distribution, the data was transformed to include both a binary approval outcome (approved vs not approved) and an approval rate representing the proportion of approvals in comparison to the amount of permit requests.

4.2 Variable Assumption Testing

Approval rate is a continuous variable and can be assumed to be normally distributed. A histogram was used to inspect its distribution (see Figure 5.1). The distribution appeared approximately normal with a slight positive skew. Measures of central tendency included a mean of 27.35, a median of 27.02 and a mode of 33.33. These values indicate that, despite the approximate normality, the distribution is positively skewed, as reflected in the relatively high modal value.

Permit status is a categorical variable. Each permit can be assigned to only one status at any given time, satisfying the assumption of mutually exclusive categories required for chi-square testing.

This feature, as well as data cleaning to remove missing values for this field, made the variable suitable for categorical data analysis.

4.3 Data Handling

During the initial extraction, several structure types had zero approved permits, causing the SQL aggregates for the approval counts and rates to return NULL values. To ensure consistency across categories, the SQL queries were modified to replace null values with zero. This adjustment ensured that structure types with no recorded approvals remained in the dataset and could be meaningfully compared.

Additionally, postal codes were joined from the address table, allowing location-based analysis despite the absence of ward-level identifiers. The final dataset used for modeling and visualization included postal code, permit status, structure type, and approval indicators.

4.4 Trends in the Data

Several trends were observed when examining the dataset. The dataset is highly imbalanced across structure types. Certain categories, such as office buildings, apartment buildings, multiple-unit residences, and retail stores, appear very frequently, reflecting their dominance in Toronto's development landscape. In contrast, many other categories appear only once or a few times, representing highly specialized or infrequent building types. This imbalance suggests that development activity is focused on large-scale commercial and residential patterns.

Permit activity was also unevenly distributed across postal codes. Certain postal regions appeared frequently in the dataset, representing areas with high construction activities, while others appeared infrequently. This pattern suggests that development pressure is focused on specific parts of the city.

5 Data Visualization

5.1 The Distribution of Approval Rate by Postal Code (Histogram)

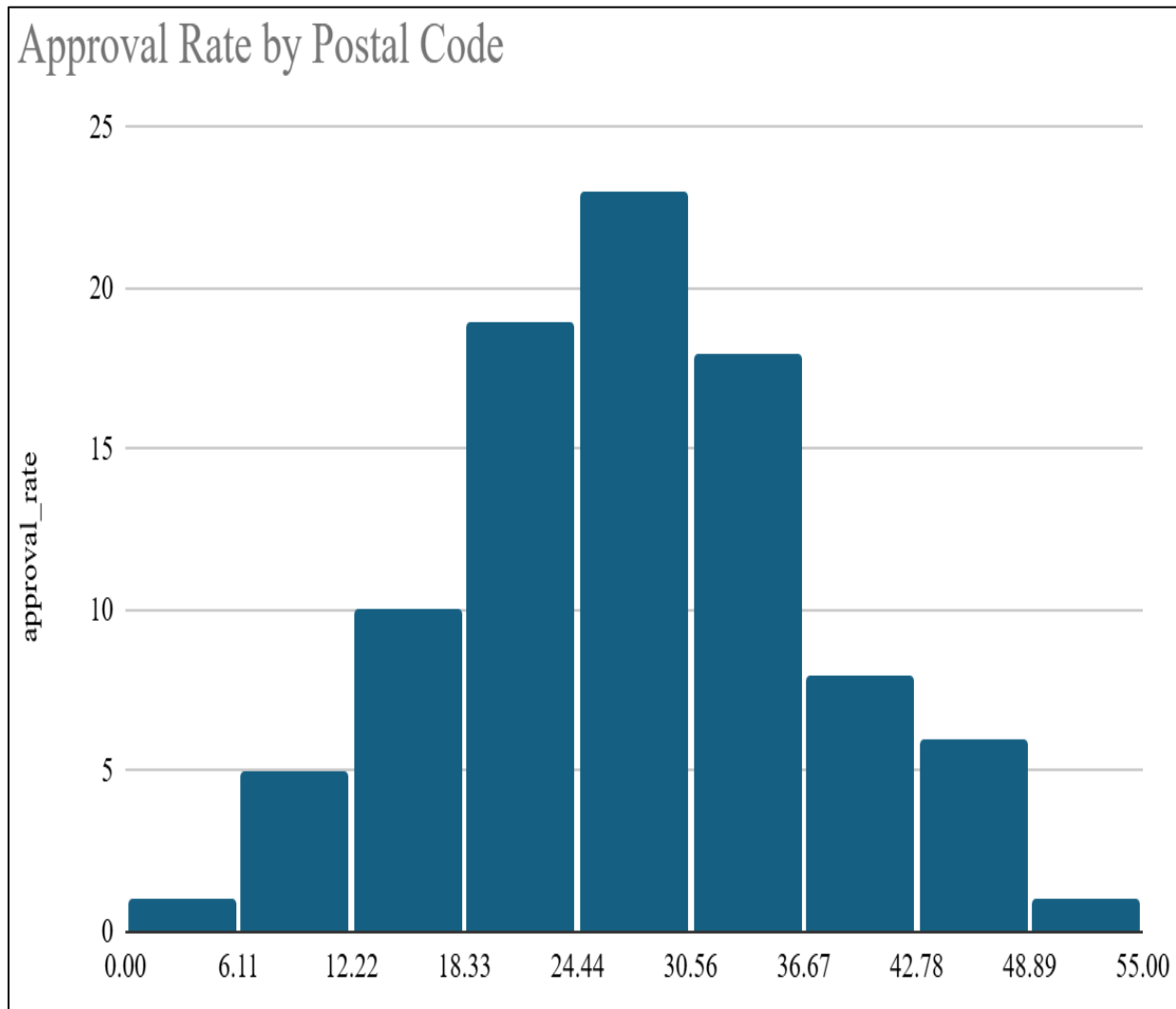


Figure 1 - Distribution of Approval Rate by Postal Code

This histogram illustrates how building permit approval rates are distributed across different postal code areas in Toronto. The distribution shows that most postal codes fall near the middle approval-rate range; however, some fall significantly above or below the median value. Overall this histogram suggests that approval rates are moderate for many areas however the presence of several postal codes that deviate from this pattern indicates that the approval outcomes may vary by location. These deviations support the need for further statistical testing to investigate whether location plays a significant role in determining permit approval outcomes.

5.2 The Approval Rate based on Structure Type (Clustered Bar Chart)

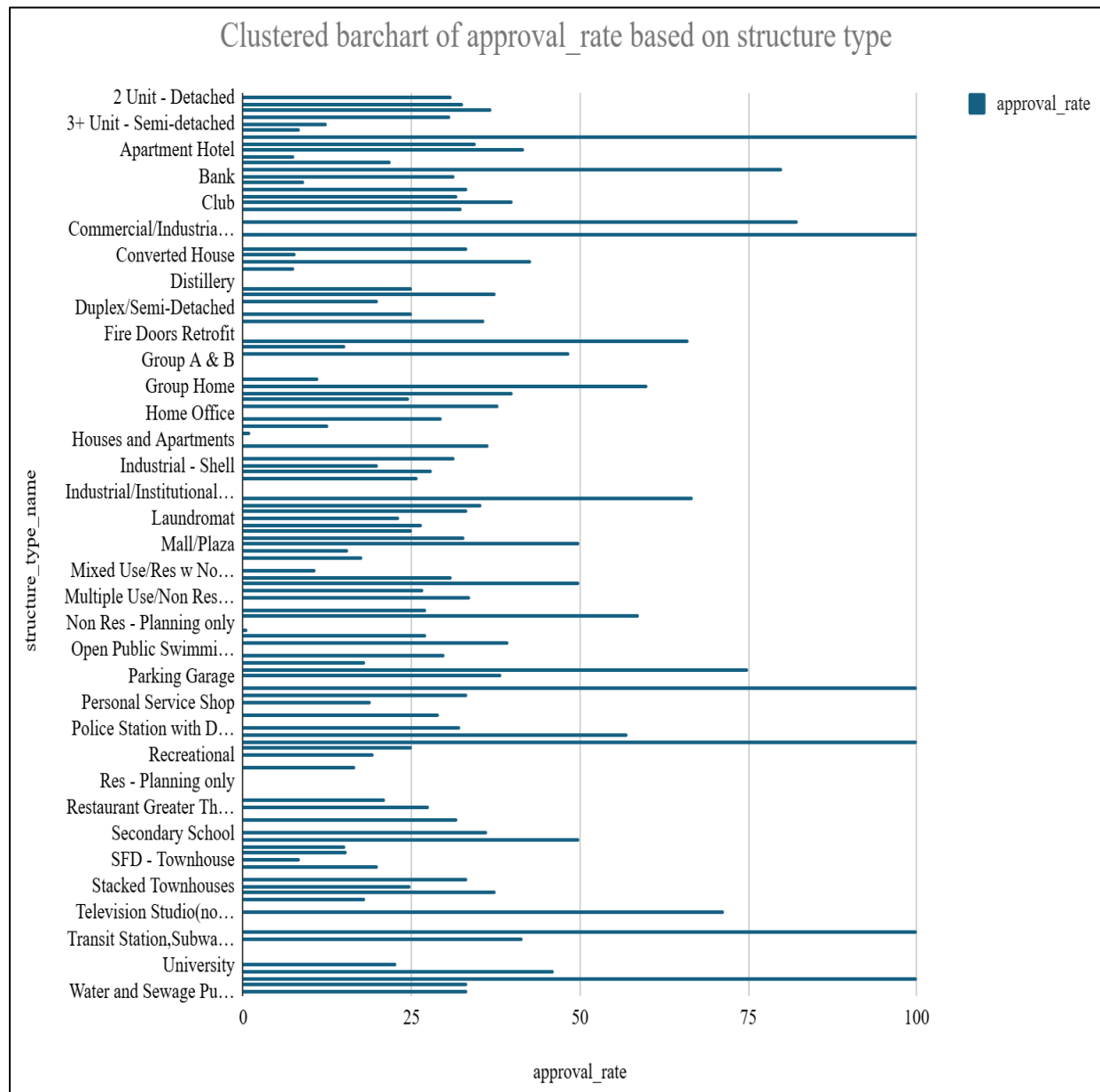


Figure 2 - Approval Rate based on Structure Type

This bar chart compares approval rates across various structure types included in the dataset. Each bar represents a single structure category, allowing for a clear comparison of how frequently different types of projects receive approval. There is significant variation among structure types. Some categories show higher approval rates, suggesting that they experience fewer regulatory barriers or a more straightforward review process. Other categories display lower approval rates, which may indicate stricter zoning requirements or more frequent application deficiencies.

5.3 The Approval Count vs Total Permits (Scatter Plot)

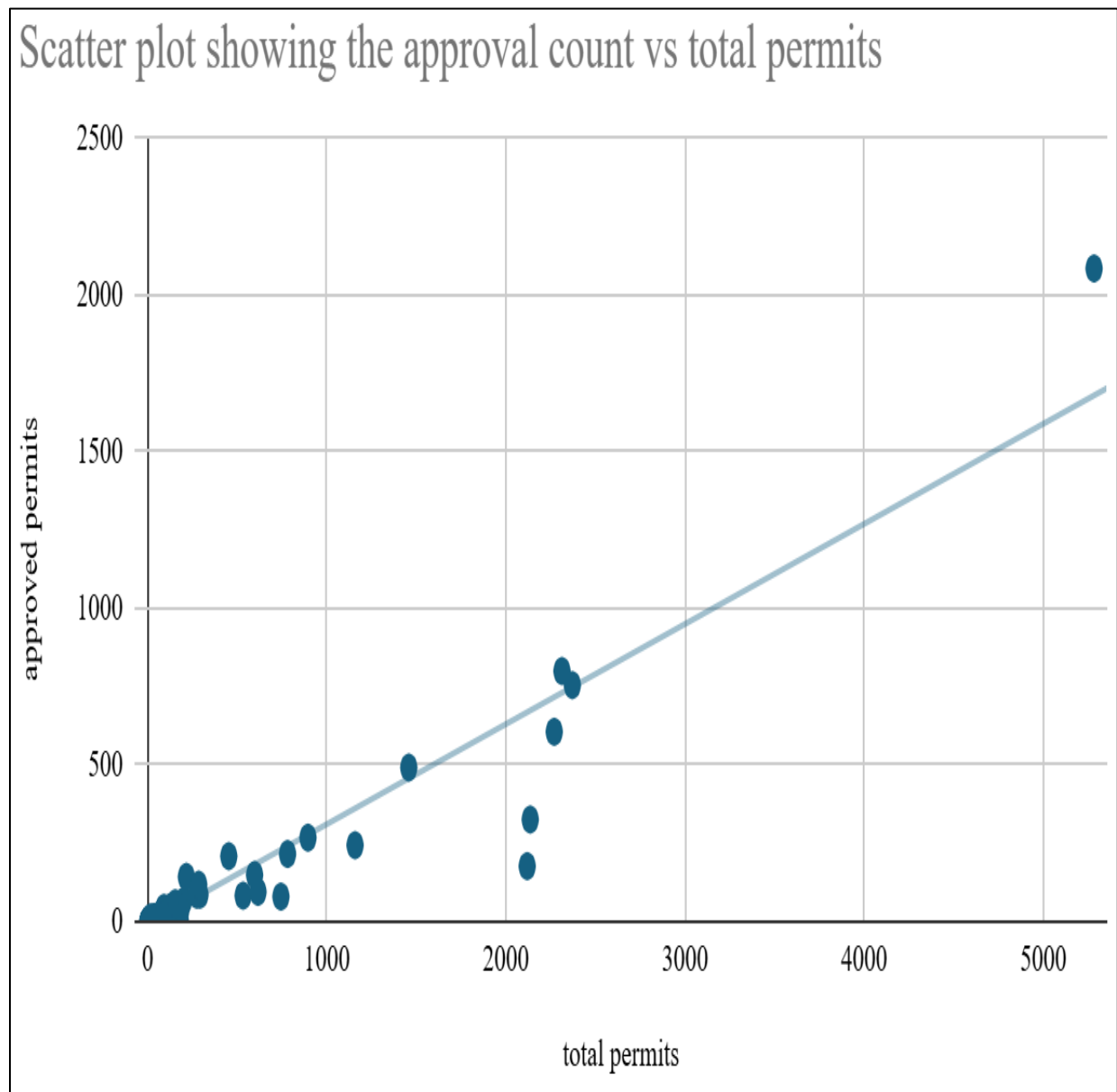


Figure 3 - Approval Count vs Total Permits

This scatter plot illustrates the relationship between the number of permits submitted and the number of permits approved across different structure types. The data points demonstrate a clear upward trend, where more application requests correlate with more approved requests. Common structure types such as office buildings and apartment complexes appear more frequently in the dataset, resulting in disproportionately high approval counts. This pattern demonstrates how project volume influences approval outcomes and suggests that certain structure types receive greater prioritization within Toronto’s permitting process.

5.4 *H1: The Approval Rate by Top 15 Postal Codes (Bar Chart)*

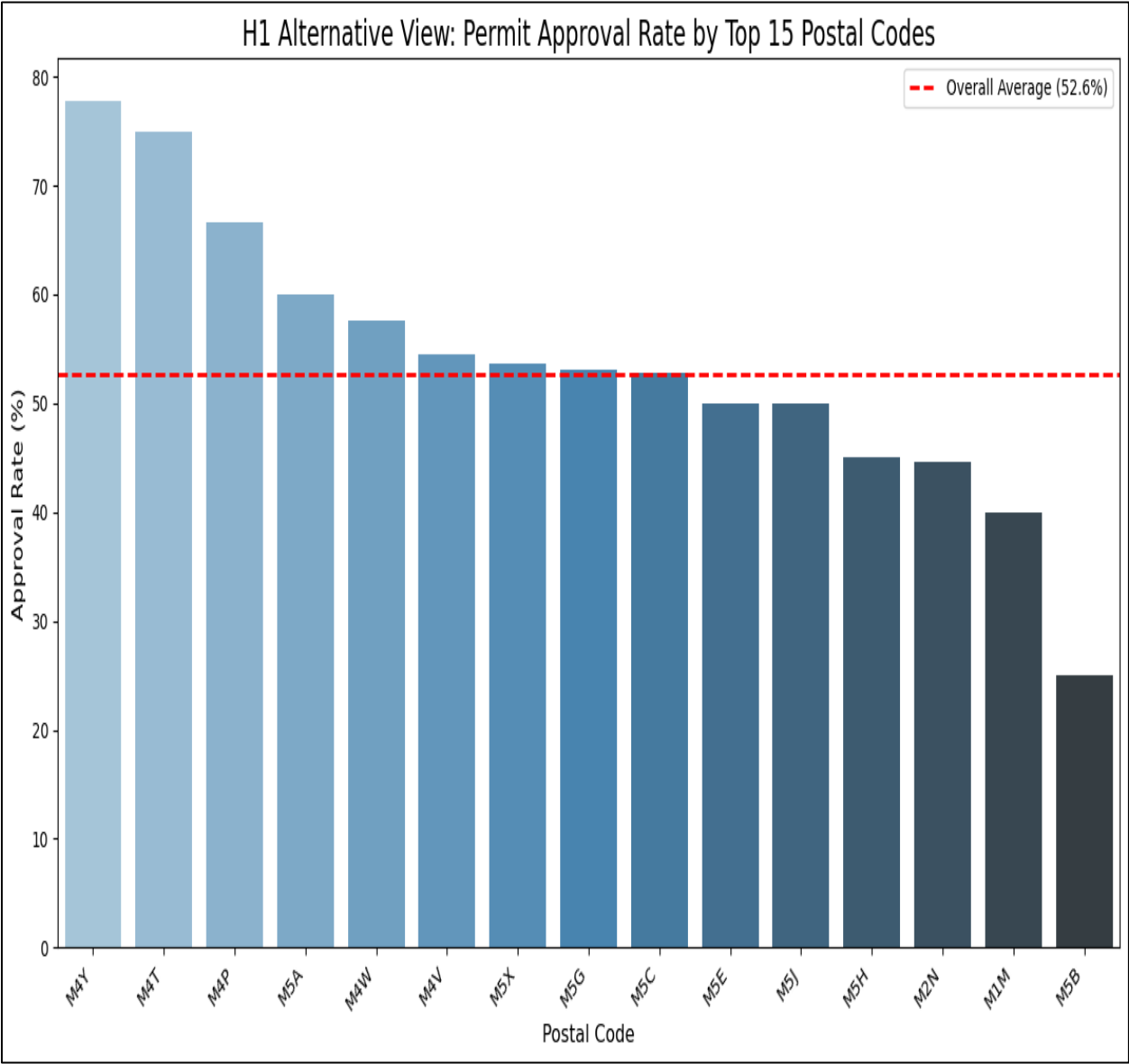


Figure 4 - H1: Approval Rate by Top 15 Postal Codes

This bar chart presents permit approval rates across the top 15 postal codes and highlights how approval outcomes vary by location. Several postal codes significantly exceed the overall average approval rate of approximately 52.6%, while others fall well below it, creating a clear downward trend from left to right. The highest-performing areas show approval rates near 80%, whereas the lowest drop to around 25%, indicating substantial geographic disparities. This pattern suggests that postal code plays an important role in permitting outcomes, with some regions experiencing consistently smoother approval processes than others.

5.5 H2: The Permit Status Distribution by Top 10 Postal Codes (Stacked Bar Chart)

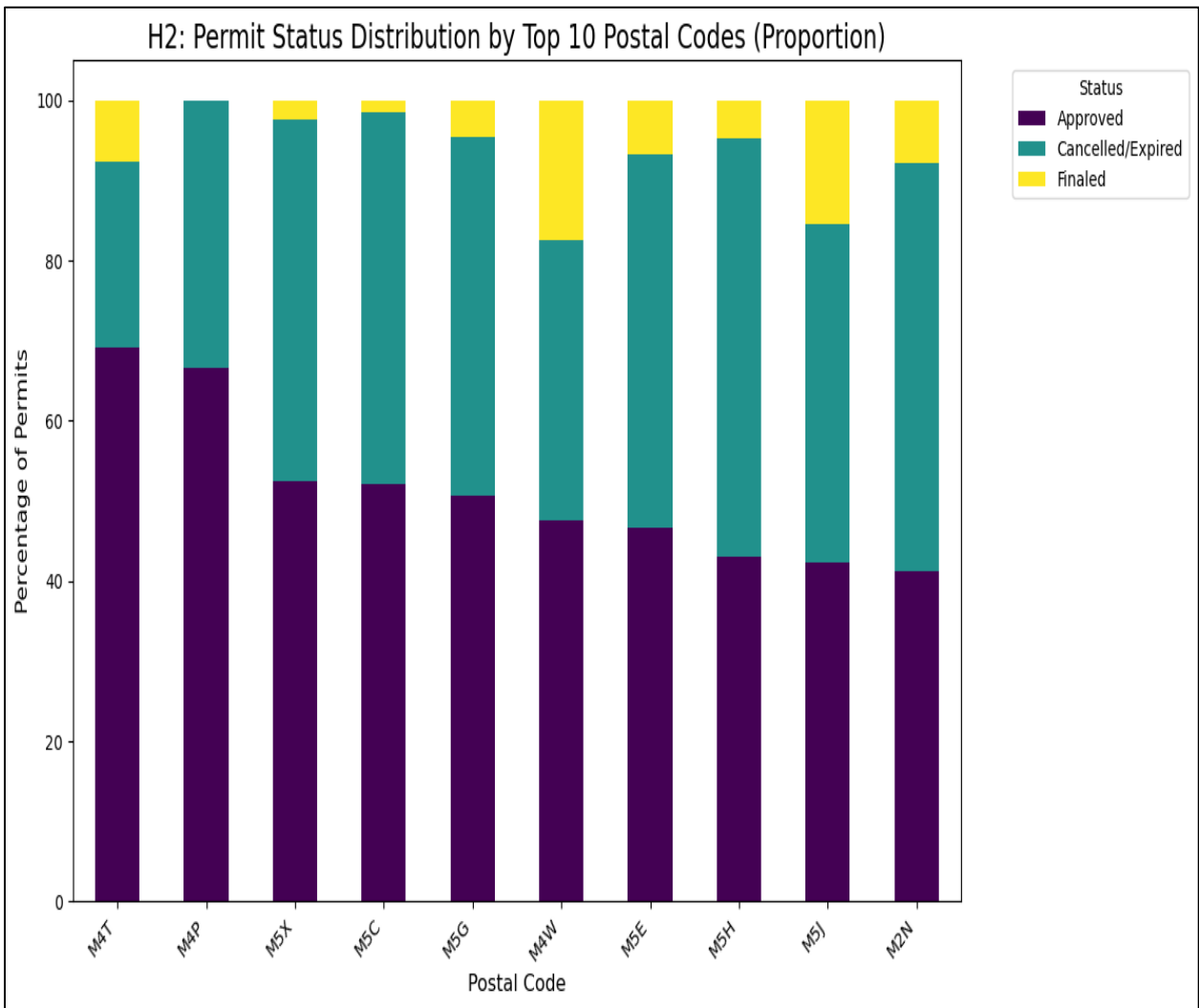


Figure 5 - H2: Permit Status Distribution by Top 10 Postal Codes

This stacked bar chart shows the distribution of permit outcomes across the top 10 postal codes, breaking requests into three categories: approved, cancelled/expired, and finalized. Most postal codes have a majority of approved permits, generally ranging between 40% and 70% of all applications. A substantial share of permits are cancelled or expired in every postal code, indicating consistent attrition in the application process. Finalized permits make up only a small proportion across all areas, suggesting that relatively few projects reach full completion compared to the number initially submitted. Overall, the chart highlights how approval dominance varies geographically while also revealing that cancellations represent a significant portion of permitting activity across Toronto’s most active postal zones.

5.6 H3: The Permit Approval Rate by Structure Type (Bar Chart)

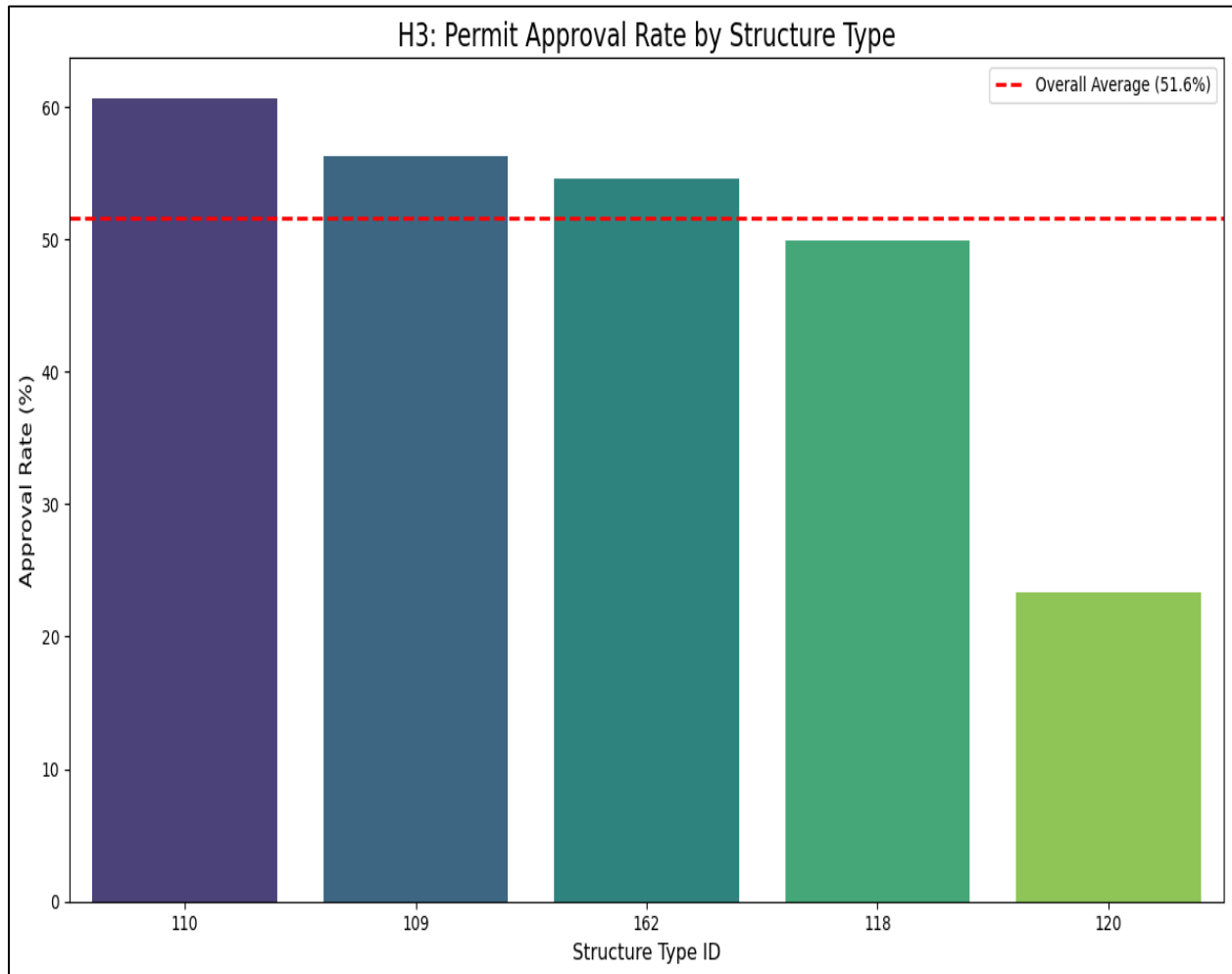


Figure 6 - H3: Permit Approval Rate by Structure

This bar chart displays the permit approval rates for different structure types compared to the overall average approval rate of 51.6%. Structure Type 110 demonstrates the highest approval success at approximately 60%, significantly exceeding the overall average. Structure Types 109 and 162 also perform above average with approval rates around 56% and 54% respectively, while Type 118 hovers near the citywide average at roughly 50%. However, Structure Type 120 stands out with a notably lower approval rate of approximately 23%, falling well below the average and suggesting potential regulatory challenges or stricter requirements for this particular structure classification. These variations indicate that structure type plays a meaningful role in permit approval outcomes, with certain categories experiencing considerably different success rates than others.

6 Model Building

6.1 Model Development and Justification

The primary goal of the modelling stage was to test whether the postal code and structure type in the dataset were statistically associated with permit approval outcomes. The chi-square test of independence was the most appropriate analytical technique for hypothesis 2, as both the independent and dependent variables are categorical variables, the sample size is large, and the expected cell counts are sufficient for reliable inference. The model was constructed to evaluate three separate hypotheses.

For Hypotheses 1 and 3, although a chi-square test could also assess simple association, a more suitable method was needed because the research question shifted toward predicting a binary outcome:

- **Permit Approval (1 = Issued, 0 = Not Issued).**

To support this predictive objective, Logistic Regression was used. This model quantifies the predictive power of each independent variable and estimates how strongly location or structure type influences the probability of approval.

A significance threshold of $\alpha=0.05$ was used throughout the analysis. For the Chi-Square test (H2), when the p-value fell below this threshold, the null hypothesis of independence was rejected, confirming the existence of a statistically significant correlation. For the Logistic Regression models (H1 and H3), the predictive success was primarily evaluated using the ROC AUC (Area Under the Receiver Operating Characteristic Curve), a robust measure of model discrimination, alongside the traditional classification metrics and coefficient p-values to fully justify the model's contribution.

6.2 Data Preparation for Model Implementation

Data retrieved from the SQL queries were saved as CSV and implemented into a Python data frame as shown in Figure 6.1. One file contained permit records with postal codes and the associated structure types for the permit. The second summarized count of permit status by postal code.

Before running the model, the postal dataset was expanded so that each permit was represented individually. This ensured that the chi-square test used true frequencies rather than aggregated category counts. A binary variable (`approved = 1` if `status_id = 32`, otherwise `approved = 0`) was created to distinguish approved permits from all the other outcomes.

After preparing the data, a contingency table was generated comparing postal codes to approval status. This table served as the input for the chi-square model. This model allowed for the examination of both categorical variables effectively to determine whether a statistically meaningful correlation existed.

6.3 Model Implementation

The analytical approach differed depending on whether the task involved correlation or prediction.

- **Chi-Square Test:** Conducted using `scipy.stats.chi2_contingency()` to examine the relationship between postal codes and the full set of permit statuses (H_2).
- **Logistic Regression:** Implemented using `sklearn.linear_model`. Logistic Regression to determine whether postal codes (H_1) or structure type IDs (H_3) could predict approval outcomes.

During the first test, H_1 , postal codes were One-Hot Encoded and used as independent variables to predict the binary approval variable (`is_approved`). Logistic Regression evaluated whether location added predictive value.

The second test, H_2 , evaluated the overall relationship of location to the full set of permit statuses multi-class `status_id`. The appropriate statistical technique for such a test-the Chi-Square test of independence-required a contingency table of location against full status set.

The third test, H_3 , used Logistic Regression in a similar way to test whether structure types, or `structure_type_id`, could predict binary approval rates.

Each model followed a similar structure: preparation of input variables, the application of the statistical test/model, and recording the key output metrics. To interpret the performance of the Logistic Regression models, Confusion Matrices and ROC Curves were used for visualizations,

while in the case of the Chi-Square test, regional differences in the distribution of outcomes were assessed through the use of a Stacked Bar Chart.

6.4 *Model Output*

Each test produced a set of key outputs tailored to the methodology used:

- **Chi-Square Test (H₂):** The outputs included a χ^2 statistic, degrees of freedom (DOF), and a P-value, which were used to evaluate the correlation and determine whether the null hypothesis could be rejected against the $\alpha=0.05$ threshold.
- **Logistic Regression (H₁ & H₃):** The main outputs consisted of the ROC AUC score, which measured the model's overall ability to discriminate between approved and non-approved permits, along with the Classification Report containing Accuracy, Precision, and Recall. Coefficient significance was also reviewed to determine whether each variable contributed meaningfully to prediction. Together, these results indicated how effectively postal codes and structure types predicted approval outcomes.

7 Model Evaluation

7.1 Model Performance

The chi-square test of independence and logistic regression models were evaluated to determine whether location and structure type were statistically related to permit approval outcomes. Model performance was assessed using each method's standard diagnostic outputs, including chi-square values, degrees of freedom, p-values, ROC AUC scores, and coefficient significance. A significance threshold of $\alpha = 0.05$ was applied across all tests.

7.2 Evaluation of Hypothesis 1

This test was conducted to determine if there is a significant relationship between a project's Postal Code (representing the location) and the approval status of the permit.

- **Chi-square value:** 1068.6858791853758
- **Degrees of freedom:** 83
- **p-value:** 5.39×10^{-171}

Regression Result:

Variable	Coefficient	Standard Error	z	P > z	0.025	0.975
constant	-0.4055	0.913	-0.444	0.657	-2.195	1.384
P_M2N	0.1919	0.959	0.200	0.841	-1.687	2.071
P_M4P	1.0986	1.065	1.032	0.302	-0.988	3.185
P_M4T	1.5041	1.027	1.464	0.143	-0.510	3.518
P_M4V	0.5878	1.095	0.537	0.592	-1.559	2.735

P_M4W	0.7108	0.978	0.726	0.468	-1.207	2.629
P_M4Y	1.6582	1.215	1.365	0.172	-0.723	4.040
P_M5A	0.8109	1.291	0.628	0.530	-1.719	3.341
P_M5B	-0.6931	1.472	-0.471	0.638	-3.578	2.192
P_M5C	0.5199	0.944	0.551	0.582	-1.330	2.370
P_M5E	0.4055	1.058	0.383	0.702	-1.668	2.479
P_M5G	0.5292	0.918	0.577	0.564	-1.270	2.328
P_M5H	0.2081	0.931	0.224	0.823	-1.616	2.032
P_M5J	0.4055	1.008	0.402	0.687	-1.569	2.380
P_M5S	1.0986	1.528	0.719	0.472	-1.895	4.093
P_M5X	0.5521	0.965	0.572	0.567	-1.339	2.444
P_M8Y	0.4055	1.683	0.241	0.810	-2.894	3.705

The chi-square test results showed a very strong statistical association between postal code and permit approval ($\chi^2 = 1068.69$, $df = 83$, $p \approx 5.39 \times 10^{-171}$), leading to rejection of the null hypothesis at the 0.05 level. This indicates that location and approval status are not independent across the dataset. However, when postal codes were incorporated into a logistic regression model through one-hot encoding, none of the location variables produced statistically significant coefficients; all p-values exceeded 0.05. This means the model failed to reject the null hypothesis and suggests that postal code does not meaningfully improve the prediction of whether an individual permit will be approved. The contrast between the two methods indicates that the chi-square test detected broad

distributional differences across regions, but once the data were analyzed within a predictive framework that accounts for variance and individual-level effects, the influence of location became negligible. In practical terms, location may be statistically associated with approval outcomes at the population level, but it does not provide substantive predictive power in a regression context.

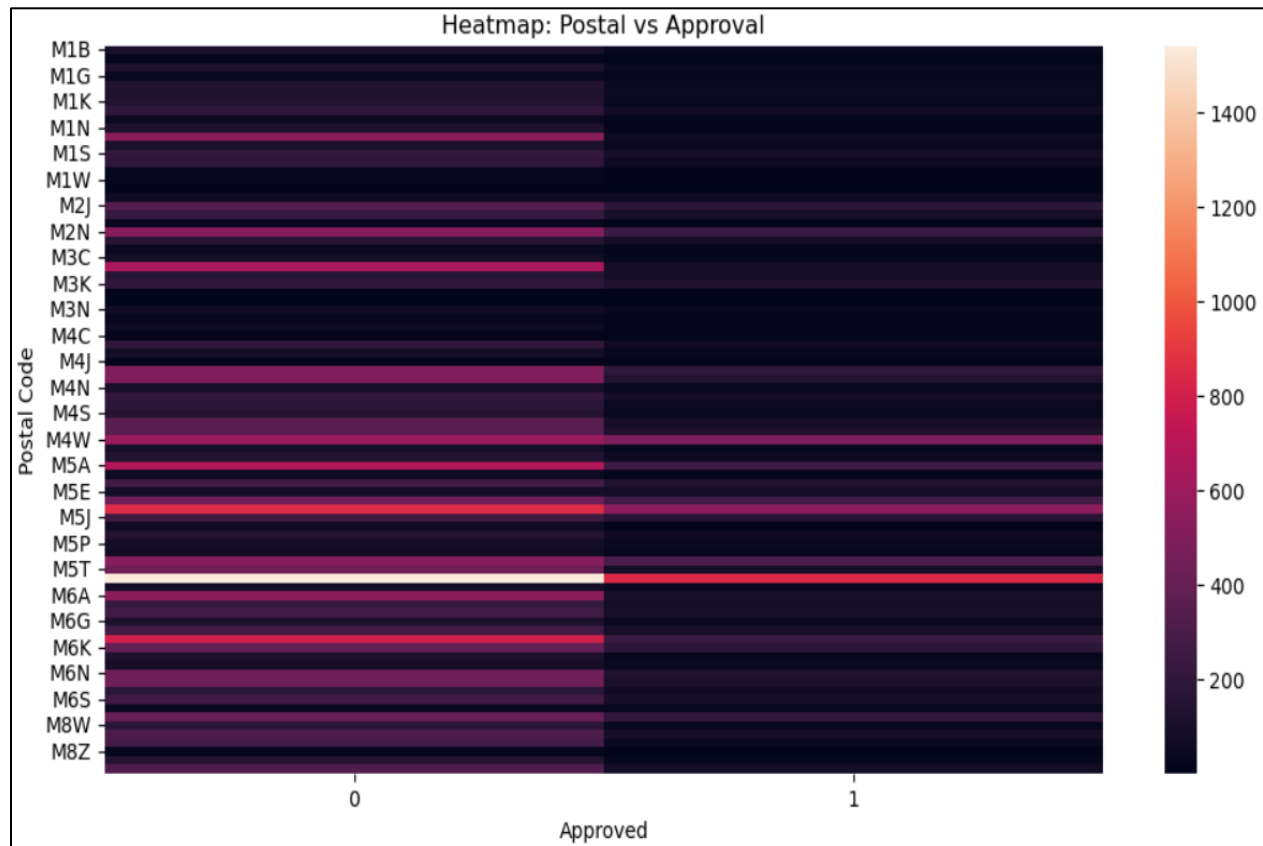


Figure 7 – Postal vs Approval by Chi Square Model (Heat Map)

7.3 Evaluation of Hypothesis 2

This test was conducted to determine if there is a significant relationship between a project's Location and the distribution of all potential permit statuses throughout the application process.

- **Chi-square value:** 4756.674164708941
- **Degrees of freedom:** 594
- **P-value:** ≈ 0.0000
- **Decision:** Reject H_0 .

Hypothesis 2 investigated whether postal code is associated with the distribution of all permit statuses throughout the application process. The chi-square test produced a highly significant result ($\chi^2 = 4756.67$, $df = 594$, $p \approx 0.0000$), leading to rejection of the null hypothesis.

This outcome indicates that postal code is strongly related to how applications progress through the entire permit lifecycle, not only to the final decision. This relationship may reflect geographic differences in administrative workflows, variations in application quality, or differing regulatory or zoning constraints across postal areas.

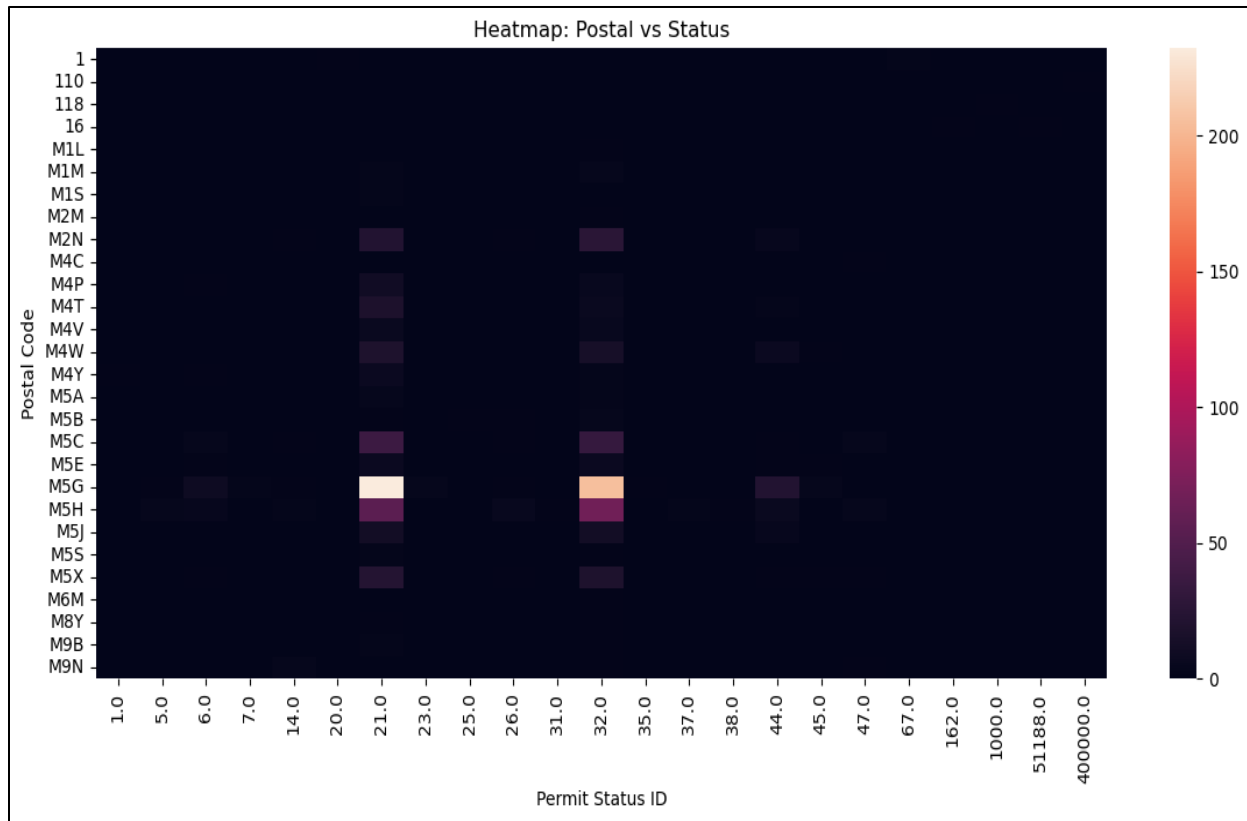


Figure 8 - Postal vs Status by Chi Square Model (Heat Map)

7.4 Evaluation of Hypothesis 3

This test was conducted to determine if there is a significant relationship between a project's structure type and the final approval status of the permit.

- **Chi-square value:** 68.57957300045224
- **Degrees of freedom:** 41

- **p-value:** 0.004429275662442323
- **Decision:** Reject H_0

Regression Result:

Variable	Coefficient	Standard Error	z	P > z	0.025	0.975
constant	0.2513	0.356	0.705	0.481	-0.447	0.950
S_110	0.1815	0.402	0.452	0.651	-0.606	0.969
S_118	-0.2559	0.369	-0.693	0.488	-0.979	0.467
S_120	-1.4409	0.560	-2.574	0.010	-2.538	-0.344
S_162	-0.0690	0.400	-0.172	0.863	-0.854	0.716

The type of structure (e.g., residential, commercial, or industrial) is statistically related to whether a permit is classified as Approved or Not Approved.

The chi-square test for the third hypothesis showed a statistically significant association between structure type and permit approval ($\chi^2 = 68.58$, $df = 41$, $p = 0.0044$), leading to rejection of the null hypothesis at the 0.05 level. This indicates that approval outcomes vary across residential, commercial, and industrial structures, likely reflecting differences in regulatory requirements and review complexity.

The logistic regression analysis further clarified which structure categories influenced approval probability. Among all structure types, S_120 displayed a statistically significant negative coefficient ($p = 0.010$), meaning permits in this category were less likely to be approved. This finding is consistent with Ontario building permit guidelines, which explain that more complex building types require additional review steps that can reduce approval likelihood. Overall, structure type plays a meaningful role in approval outcomes, supported by both the chi-square results and the regression model.

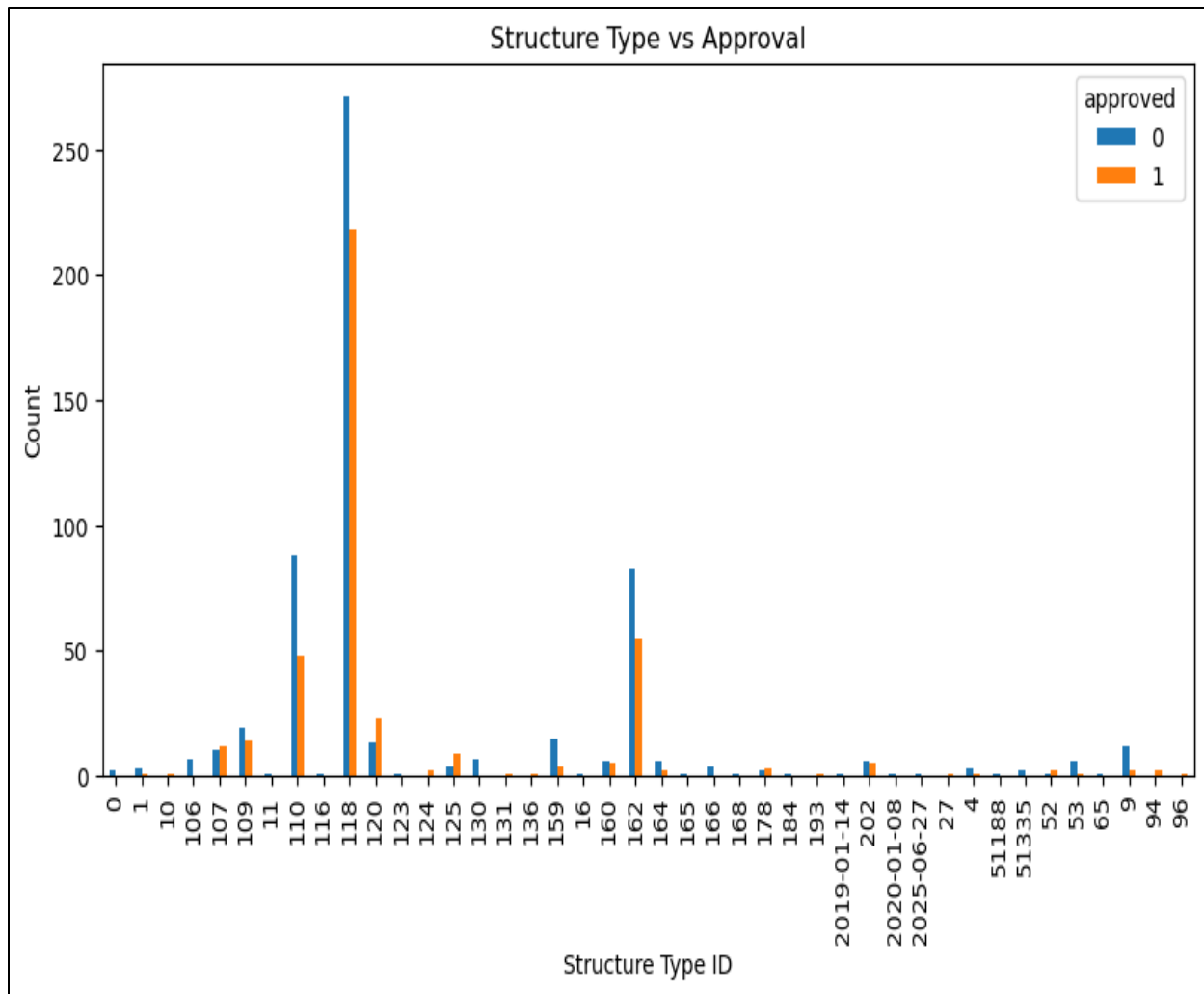


Figure 9 - Structure Type vs Approval by Chi Square Model

7.5 Limitations

The chi-square models produced meaningful insights; however, several limitations were considered when implementing the model. Firstly, some postal codes and structure types appear far more frequently than others, which can contribute to the significance. Additionally, the chi-square test only indicates whether there is a relationship between the variables and does not quantify its strength. Finally, it does not account for the external factors such as policy changes or staffing differences between districts, which may influence permit decisions.

Conclusions and Recommendations

8.1 Conclusion

For this study we set out to answer 3 research questions based on the available information. The research questions: ‘Does the geographic location affect the likelihood of a building permit being approved?’, ‘To what extent does structure type, along with its location, influence approval outcomes?’ and ‘Are there observable biases in permit approvals across different structure type?’, guided our exploration of the available data. The answers of which will assist with improving the permit approval system in the Toronto area. Which in turn yielded the following results and insights.

a. Does the geographic location affect the likelihood of a building permit being approved?

- **Hypothesis Set 1: Location and Permit Category**

H₀ (Null Hypothesis): There is no correlation between location and permit approval.

H₁ (Alternative Hypothesis): There is a significant correlation between location and permit approval

The Chi-square statistical model employed yielded a chi-square value of 1068.6858 and a p-value of 5.39×10^{-171} , leading to the rejection of the null hypothesis, proving that there is a statistically significant correlation between location (postal code) and permit approval. The regression showed that the correlation between location and permit approval was statistically insignificant. This is because the regression generated p-values all exceeded 0.05 which would lead us to fail to reject the null hypothesis. However, when these same variables were entered into a logistic regression model, the effect of location was not statistically significant. This difference suggests that although the chi-square test detected an overall correlation, the effect size was small and became non-significant once model-based controls and parameter estimates were considered. In other words, location does not meaningfully predict approval probability after accounting for other factors. This means that independently the chi squared test would have produced a type of error. It can be surmised that other factors strongly correlated with location would lead to the strong correlation between location and permit outcome, such as access to resources and education.

b. To what extent does structure type, along with its location, influence approval outcomes?

- **Hypothesis Set 2: Location and Permit Status**

H₀ (Null Hypothesis): There is no correlation between project location and permit status.

H₁ (Alternative Hypothesis): There is a significant correlation between project location and permit status.

The scatter plot (Figure 2) displays an observable pattern between postal codes and approval rates, suggesting a potential relationship. This visual trend was confirmed through the chi-square test, which produced a chi-square value of 4756.6741 and a p-value effectively equal to 0.0000, well below the 0.05 significance threshold. As a result, the null hypothesis was rejected, indicating a statistically significant association between location and permit status. This relationship is further reinforced in the heat map (Figure 5), where distinct clusters and color variations highlight areas with higher approval concentrations. The strength of the statistical significance also implies a low likelihood of a Type I error. These findings suggest that geographic factors may influence where permits progress successfully through the approval process, potentially reflecting differences in local resources, support networks, or access to information across postal code areas.

c. Are there observable biases in permit approvals across structure type?

- **Hypothesis Set 3: Structure Type vs Permit Status**

H₀ (Null Hypothesis): There is no correlation between project structure type and permit approval.

H₁ (Alternative Hypothesis): There is a significant correlation between project structure type and permit approval.

The chi-square test showed a statistically significant relationship between structure type and permit approval outcomes ($\chi^2 = 68.5795$, $p = 0.004429$), leading to rejection of the null hypothesis and indicating that approval likelihood varies across different building categories. Logistic regression added further detail by identifying that structure type S_120 had a significantly negative coefficient ($p = 0.010$), suggesting it is less likely to be approved due to the added complexity associated with this building type. In contrast, several categories, such as air-supported structures, commercial/institutional buildings, penthouse/mechanical rooms, and powerplants, displayed 100% approval rates, likely reflecting the involvement of highly specialized project teams and

stronger application submissions (Trinidad & Riso, 2025). Overall, the findings demonstrate that structure type meaningfully influences permit approval outcomes, with differences likely tied to varying levels of project complexity, regulatory requirements, and the expertise of those preparing the applications.

8.2 Recommendation

a. Modify the Permit Approval Process

Given that overall approval rates remain below 30% across most locations and structure types, the system itself may require review. Inefficiencies or unclear requirements could contribute to high rejection rates independent of project characteristics.

b. Conduct Comprehensive Education and Sensitization

A comprehensive education campaign would also sensitize the populace about the requirements needed for different types of permits. The inclusion of a section with frequently asked questions and frequent reasons for permit rejections would help significantly.

c. Conduct Further Research

Further research should be done to determine the additional factors that may lead to the high approval rate in the most successful locations and structure types. How these factors can be employed in less successful areas and structure types, as more permit approvals would lead to more development, which would assist with the development and prosperity of the greater Toronto area.

References

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- Kamellia, S., McCabe, B. Y., Shahi, A., De Berardis, P., & Lyall, R. (2017). Evaluation of tall building construction permitting process in Toronto. In *Leadership in sustainable infrastructure* (pp. CON124–1–CON124–10). Building SMART Canada https://legacy.csce.ca/elf/apps/CONFERENCEVIEWER/conferences/2017/pdfs/CONSPEC/FinalPaper_124.pdf
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- Trinidad, J., & Riso, T. (2025, June 6). *Construction permitting: Essential steps for compliance and approval*. Procore. <https://www.procore.com/library/construction-permitting>

Appendix

SQL Codes:

3.1

Total number of Permits

```
SELECT
    COUNT(*) AS total_permits
FROM Permit;
```

List all tables

```
SHOW TABLES;
```

Permit table

```
SELECT *
FROM Permit;
```

Permits

```
SELECT
    P.permit_id, a.postal, p.status_id, p.structure_type_id, p.est_const_cost
FROM Permit p
JOIN Address a ON p.address_id = a.address_id;
```

3.2

Number of Permits for each structure type

```
SELECT
    st.structure_type_name, COUNT(*) AS count_permits
FROM Permit p
JOIN Structure_Type st ON p.structure_type_id = st.structure_type_id
GROUP BY st.structure_type_name
ORDER BY count_permits DESC;
```

#Structure Type vs Permit Status

```
SELECT
    st.structure_type_name, COUNT(*) AS total,
    COALESCE(SUM(CASE WHEN ps.status_name = 'Permit Issued' THEN 1 ELSE 0 END), 0)
    AS approved_count,
    COALESCE((SUM(CASE WHEN ps.status_name = 'Permit Issued' THEN 1 ELSE 0 END)
    / COUNT(*)) * 100, 0) AS approval_rate
FROM Permit p
JOIN Structure_Type st ON p.structure_type_id = st.structure_type_id
JOIN Permit_Status ps ON p.status_id = ps.status_id
GROUP BY st.structure_type_name
ORDER BY st.structure_type_name, total DESC;
```

3.3

```
SELECT
    postal, COUNT(*) AS total_permits, SUM(CASE WHEN ps.status_name = 'Permit Issued'
    THEN 1 END) AS approved_count, (SUM(CASE WHEN ps.status_name = 'Permit Issued'
    THEN 1 END) / COUNT(*)) * 100 AS approval_rate
FROM Permit p
JOIN Address a ON p.address_id = a.address_id
JOIN Permit_Status ps ON p.status_id = ps.status_id
GROUP BY postal
ORDER BY approval_rate DESC;
```

6.1

```
df = pd.read_csv("permit_table_with_postal.csv")
df_postal = pd.read_csv("postal_table.csv")

def chi_square_test(table, var1, var2):
```

```

chi2, p, dof, expected = stats.chi2_contingency(table)

print(f"\n=== Chi-Square Test: {var1} vs {var2} ===")
print("Chi-square value:", chi2)
print("Degrees of freedom:", dof)
print("p-value:", p)

if p < 0.05:
    print("➔ Conclusion: Reject H0 — There IS a significant relationship.\n")
else:
    print("➔ Conclusion: Do NOT reject H0 — No significant relationship.\n")

```

6.2

```

# Crear la columna approved
df["approved"] = (df["status_id"] == 32).astype(int)

# Crosstab
ct1 = pd.crosstab(df["postal"], df["approved"])

# Run chi-square
chi_square_test(ct1, "postal", "approved")

# Heatmap
plt.figure(figsize=(12,6))
sns.heatmap(ct1, annot=False)
plt.title("Heatmap: Postal vs Approval")
plt.xlabel("Approved")
plt.ylabel("Postal Code")
plt.show()

```